MAHATMA GANDHI UNIVERSITY, KOTTAYAM

BSc (Honours) Computer Science

THIRD SEMESTER EXAMINATION

(2024 ADMISSION ONWARDS)

MG3DSECSC200: INTRODUCTION TO DATA SCIENCE

MODEL QUESTION PAPER

Duration: 2 Hrs Maximum Marks: 70

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A: Very Short Answer Questions.

Answer All Questions. Each question carries 2 marks.

1. Differentiate between a data scientist and a data analyst. [U][1]
2. List two key activities performed during the data preparation stage. [K][1]
3. Distinguish between dimensionality reduction and numerosity reduction. [U][2]
4. List any two Python libraries used for EDA. [K][3]
5. Discuss the key advantage of Leave-One-Out Cross-Validation (LOOCV). [U][4]

(5 X 2 = 10 Marks)

Part B: Short Answer Questions.

Answer any 6 out of 8 Questions. Each question carries 5 marks

6. Compare the objectives of data preparation stage with modelling stage. [U][1]
7. List and explain the types of plots commonly used in EDA with examples. [A][3]
8. Summarize the role of data visualization in the steps of EDA. [U][3]
9. Explain the concept of Simple Linear Regression with an example. [U][4]
10. Describe cross-validation. How does k-fold cross-validation improve model evaluation? [An][4]
11. Create a small dataset of atleast 10 instance with actual and predicted values. (a)Calculate MSE(b)Analyze with predictions. [A][4]
12. What is the benefit of using more folds in cross-validation? [U][4]
13. Illustrate the features of data collected through crowdsourcing. [U][2]

(6X 5 = 30 Marks)

Part C: Essay Questions.

Answer any 2 out of 3 Questions. Each question carries 15 marks

14. Illustrate the steps involved in a data science modelling. [U][4]

15. Define the following terms and provide a simple example of each: (i) Data Discretization, (ii) Data Reduction, (iii) Binning, (iv) Normalization. [U][2]

16. Explain in detail about various tools used for performing EDA. [U][3]

(2 X 15 = 30 Marks)